INCH-POUND

A-A-60049

28 November 2022

COMMERCIAL ITEM DESCRIPTION

RODS, THREADED, CADMIUM AND HEXAVALENT-CHROMIUM FREE

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1 SCOPE. This commercial item description (CID) covers rods which shall not contain cadmium or hexavalent-chromium in their finishes.

2 CLASSIFICATION

2.1 Types.

Type I - AIA/NAS1454 - Rod, Continuous Thread

3 SALIENT CHARACTERISTICS.

3.1 Materials. Rods shall be manufactured as specified in the Aerospace Industry Association (AIA) documents listed in 2.1 by Type, except as noted, and the plating/finishes shall not contain cadmium and hexavalent-chromium as specified herein.

3.2 Dimensions. Dimensions for all designated types of rods apply after plating/finishing and prior to any application of lubricants (if any). Refer to each individual NAS/NASM specification listed in Section 2.1 for detailed part geometry and tolerance requirements.

3.3 Hexavalent-chromium free PIN codes and finish/plating options. Finish/plating options for each type of rod shall be specified in the contract. PIN codes and finish/plating options for all designated as specified in the contract for the rods are:

P5 = CRES rods shall be cleaned, descaled and passivated in accordance with SAE AMS2700 Method 1, Type 6, 7, or 8, or Method 2; or ASTM A967/A967M. Passivation treatments containing hexavalent chromium are prohibited.
QW = Carbon steel rods shall be Zi-Ni plated in accordance with MIL-PRF-32647, Type K20.
QX = Carbon steel rods shall be Zi-Ni plated in accordance with MIL-PRF-32647, Type K16.

(Uncoated brass rods are not included in this CID because do not contain hazardous materials.)

3.4 Interchangeability. When AIA/NASM specifications list interchangeability parts, do not interchange parts containing cadmium or hexavalent-chromium with the cadmium and hexavalent-chromium free parts listed herein without design authority approval.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FSC 5307

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4 REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraph 23.403, 23.404(b) of the Federal Acquisition Regulation (FAR), or Federal Sustainability contain in EO 14057; as applicable.

5 PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Responsibility for inspection. The contractor is responsible for the performance of all inspections.

5.3 Contractor certification. The contractor shall certify and maintain substantiating evidence the product offered meets the salient characteristics of this commercial item description and conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to perform or witness any of the inspections it deems necessary to ensure supplies and services conform to prescribed requirements.

6 PACKAGING. Preservation, packaging, packing, and marking shall be as specified in the contract or order.

7 NOTES

The following part or identification (PIN) numbering procedure is for government purposes and does not constitute a requirement for the contractor.

7.1 Table. The following table incorporates new finish/plating codes which are cadmium and hexavalent-chromium free for the designated rod types.

TABLE I. Type I (-1) Rod, continuous thread. /1

A-A-60049 Carbon Steel Part Number Zinc-Nickel Plated, Type K16/2	A-A-60049 Carbon Steel Part Number Zinc-Nickel Plated, Type K20/3	Carbon Steel Part Number from NAS1454	A-A-60049 CRES Part Number Passivated	CRES Part Number from NAS1454
AA60049-1-QX02-( )	AA60049-1-QW02-( )	NAS1454-02-( )	AA60049-1-P502-( )	NAS1454C02-( )
AA60049-1-QX04-( )	AA60049-1-QW04-( )	NAS1454-04-( )	AA60049-1-P504-( )	NAS1454C04-( )
AA60049-1-QX06-( )	AA60049-1-QW06-( )	NAS1454-06-( )	AA60049-1-P506-( )	NAS1454C06-( )
AA60049-1-QX08-( )	AA60049-1-QW08-( )	NAS1454-08-( )	AA60049-1-P508-( )	NAS1454C08-( )
AA60049-1-QX3-( )	AA60049-1-QW3-( )	NAS1454-3-( )	AA60049-1-P53-( )	NAS1454C3-( )
AA60049-1-QX3A-( )	AA60049-1-QW3A-( )	NAS1454-3A-( )	AA60049-1-P53A-( )	NAS1454C3A-( )
AA60049-1-QX4-( )	AA60049-1-QW4-( )	NAS1454-4-( )	AA60049-1-P54-( )	NAS1454C4-( )
AA60049-1-QX4A-( )	AA60049-1-QW4A-( )	NAS1454-4A-( )	AA60049-1-P54A-( )	NAS1454C4A-( )
AA60049-1-QX5-( )	AA60049-1-QW5-( )	NAS1454-5-( )	AA60049-1-P55-( )	NAS1454C5-( )
AA60049-1-QX5A-( )	AA60049-1-QW5A-( )	NAS1454-5A-( )	AA60049-1-P55A-( )	NAS1454C5A-( )
AA60049-1-QX6-( )	AA60049-1-QW6-( )	NAS1454-6-( )	AA60049-1-P56-( )	NAS1454C6-( )
AA60049-1-QX6A-( )	AA60049-1-QW6A-( )	NAS1454-6A-( )	AA60049-1-P56A-( )	NAS1454C6A-( )

/1 ( ) = rod length; PIN's are derived from AIA/NAS1454, for specific dimensions and material composition refer to AIA/NAS1454.
/2 ( ) = Zinc-nickel, Type K16 is a replacement for cadmium plated.
/3 ( ) = Zinc-nickel, Type K20 is a replacement for zinc with hexavalent chromium.

(Table notes on next page.)

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NOTES

PIN construction

AA60049 -1 – QW 4A 0607

- Rod Length = 6.375 inches (see AIA/NAS1454):
  - 06 = length in inches (6 inches)
  - 07 = additional length in 16ths of an inch (7 x 0.0625 = 0.4375).
- Dash Number (No "A" = UNF-2A; "A" = UNC-2A threads from AIA/NAS1454).
- Material / Finish code (see 3.3):
  - P5 = CRES / passivated
  - QW = Steel / zinc-nickel type K20 plated
  - QX = Steel / zinc-nickel type K16 plated
- Table/Type number in this CID (-1 = Table/Type I)
- CID identifier (basic part number)

Example PIN's:

AA60049-1-QW4A0412	=	a NAS1454 – Rod (all thread), carbon steel, zinc-nickel Type K20 plated, 0.2500-20, UNC-2A, 12.750-inches in length.
AA60049-1-QX404	=	a NAS1454 – Rod (all thread), carbon steel, zinc-nickel Type K16 plated, 0.2500-28, UNF-2A, 12-inches in length.
AA60049-1-P5160312	=	a NAS1454 – Rod (all thread), CRES, passivated, 1.0000-12, UNF-2A, 3.750-inches in length.

7.2 Source of government documents.

7.2.1 FAR: Copies of the Federal Acquisition Regulations are available online from https://www.acquisition.gov/browse/index/far.

7.2.2 Executive order (EO). Copies of the EO documents are available online from https://www.epa.gov/greeningepa/executive-order-13693.

7.2.3 Military specifications: Copies of military specifications are available online from https://quicksearch.dla.mil.

7.3 Sources for Non-Government documents.

7.3.1 AIA documents: Copies of Aerospace Industry Association are available online from https://www.aia-aerospace.org/standards/.

7.3.2 ASTM documents. Copies of American Society for Testing and Materials documents are available online from https://www.astm.org/.

7.3.3 SAE documents. Copies of Society of Automotive Engineers documents are available online from https://www.sae.org.

7.4 Ordering data. The contract or order should specify the following:

a. Title, number and date of this CID.
b. National stock number, (if available).
c. Material (Carbon steel or CRES).
d. Hexavalent-chromium and cadmium free part number (PIN) from this CID.
e. Quantity.

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7.5 Key words:

a. CRES
b. Carbon steel
c. Passivated
d. Plate
e. Plating
f. Steel
g. Zinc-nickel

MILITARY INTERESTS:

Custodian:
Army – AT
Navy – AS
Air Force – 71
DLA – CC

Review activities:
Army – AR, AV, CR, MI
Navy – AS, CG, MC, YD
Air Force – 03, 11, 70
DLA – GS4, GS7

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FAS

Preparing Activity:
Army – AT

(Project 5307-2023-001)

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